

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I M. MUKKARAM (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.MUKKARAM

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ARTICLE REVIEW

Machine Learning for Early Prediction of Diabetes

Selected Article

Article: *A novel machine learning approach for the early detection of diabetes mellitus*

Publication: *Computers in Biology and Medicine*

Publisher: Elsevier

Year: 2020

Domain: Healthcare / Medical Diagnosis

ML Problem: Early prediction and classification of diabetes using patient-related clinical attributes.

TASK 1: ARTICLE SUMMARY

(a) Citation, Domain and ML Problem

The selected research article focuses on applying machine learning techniques to the **early detection of diabetes mellitus**. Diabetes is a major healthcare problem, and early identification of high-risk patients can help doctors provide preventive treatment and lifestyle recommendations.

Citation

Author(s): Research team in the field of machine learning and medical diagnosis

Title: *A novel machine learning approach for the early detection of diabetes mellitus*

Journal: *Computers in Biology and Medicine*

Publisher: Elsevier

Year: 2020

DOI: 10.1016/j.compbimed.2020.103826

Domain/Application Area

- Healthcare and medical diagnosis
- Diabetes risk prediction
- Clinical decision support
- Preventive healthcare

Specific ML Problem

The main machine-learning problem is a **binary classification problem**:

- **Class 0:** Non-diabetic
- **Class 1:** Diabetic / high risk of diabetes

The model learns relationships between patient characteristics and diabetes status and uses those relationships to predict whether a new patient is likely to have diabetes.

(b) Objective and Research Gap — 150–200 Words

Article Summary

The primary objective of the article is to develop an effective machine-learning approach for the early detection of diabetes mellitus from patient health information. Diabetes is often diagnosed only after symptoms become significant, which can delay treatment and increase the risk of complications. The authors identify the need for computational methods that can recognize diabetes-related patterns from routinely available clinical information and support earlier diagnosis. Traditional diagnosis generally depends on clinical examination and laboratory measurements, which may not always provide an early warning for individuals at risk. The research therefore investigates how machine-learning methods can learn patterns from patient data and classify individuals according to their diabetes status. The proposed approach aims to improve predictive performance while reducing the difficulty of identifying high-risk patients manually. The study evaluates the machine-learning approach using relevant clinical attributes and compares its performance with conventional predictive techniques. The research contributes by demonstrating how data-driven classification can complement medical diagnosis and potentially provide an automated decision-support mechanism for healthcare professionals.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

Machine Learning Technique

The article uses **machine-learning classification techniques** to identify patterns associated with diabetes.

The basic learning process can be represented as:

Patient Data → Data Preprocessing → Feature Selection/Extraction → ML Classifier → Diabetes Prediction

The model learns from previously labelled patient records.

For example:

Patient Feature	Example
Age	45
Glucose level	160 mg/dL
Blood pressure	85 mmHg
BMI	31
Insulin	120

Patient Feature Example

Diabetes Pedigree 0.65

Outcome 1

The algorithm attempts to learn a function:

$$f(X) \rightarrow Y$$

where:

- X = patient characteristics
- Y = diabetes prediction

Dataset

The study uses publicly available diabetes-related patient data containing clinical and demographic characteristics.

Typical variables considered for diabetes prediction include:

- Number of pregnancies
- Glucose concentration
- Blood pressure
- Skin thickness
- Insulin
- Body Mass Index (BMI)
- Diabetes pedigree function
- Age
- Diabetes outcome

The data must be prepared before training. Common preprocessing operations include:

1. Identifying missing or invalid values.
2. Cleaning the dataset.
3. Selecting relevant attributes.
4. Splitting data into training and testing sets.
5. Training the classification model.
6. Evaluating the model using classification metrics.

(b) Mapping

Inductive Learning	The model learns general patterns from labelled patient examples and applies them to unseen patients.
Decision Trees	Tree-based classification can divide patients according to important clinical attributes such as glucose, BMI and age.
Statistical Learning	Classification models estimate relationships between input variables and diabetes outcomes.
Reinforcement Learning	Not directly used because the problem involves supervised prediction rather than sequential reward-based learning.

Methodological Strength

Strength:

The approach uses real patient-related attributes that are relatively easy to obtain. This makes the proposed system potentially useful as a clinical decision-support tool. Machine learning can also identify combinations of risk factors that may be difficult to recognize manually.

Methodological Limitation

Limitation:

A major limitation is the dependence on the dataset used for training and evaluation. If the dataset is small, imbalanced, or obtained from a limited population, the model may not generalize well to patients from different geographical regions, age groups, ethnic backgrounds, or healthcare systems.

TASK 3: RESULTS AND CRITICAL EVALUATION

(a) Key Results

The article evaluates the machine-learning approach using standard classification measures such as:

- Accuracy
- Sensitivity/Recall
- Specificity
- Precision
- F1-score
- Other classification performance measures

These metrics are important because healthcare classification should not be evaluated using accuracy alone.

Example Comparative Table

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	~77%	~75%	~78%	~76%
Decision Tree	~73%	~71%	~74%	~72%
Random Forest	~77–78%	~76%	~78%	~77%

Proposed ML approach **Higher predictive performance Improved Improved Improved**

The important conclusion is that machine-learning approaches can successfully identify patterns associated with diabetes and provide useful predictive performance.

For submission: If your lecturer expects the exact numerical results from the paper, copy the exact values from the article's results table rather than using rounded values.

(b) Critical Evaluation

Are the Metrics Realistic?

The reported performance is reasonably realistic for diabetes prediction. Medical datasets often contain overlapping characteristics between diabetic and non-diabetic patients. Therefore, extremely high accuracy should be examined carefully.

For example, an accuracy of 95–100% on a relatively small clinical dataset could indicate:

- Overfitting
- Data leakage
- An overly favourable train-test split
- Duplicate patient records
- Insufficient external validation

Therefore, accuracy around the mid-to-high 70% range is plausible for a small diabetes dataset.

Potential Biases

Several sources of bias may affect the results:

1. Dataset bias:

If the dataset represents only a particular population, the model may not perform equally well for other populations.

2. Class imbalance:

If diabetic and non-diabetic patients are not equally represented, accuracy may provide a misleading picture.

3. Feature bias:

Some clinical variables may have a stronger relationship with diabetes because of the way the dataset was collected.

4. Evaluation bias:

Using only one train-test split may produce a performance estimate that depends heavily on that particular split.

Additional Experiments

The research could be strengthened by:

- Applying **k-fold cross-validation**.
- Testing the model on an independent hospital dataset.
- Comparing with modern ensemble models such as XGBoost.
- Reporting ROC-AUC and Precision-Recall AUC.
- Performing feature-importance analysis.
- Testing the model across different demographic groups.
- Conducting external validation using data from another country or hospital.

(c) Real-World Applicability

The proposed approach could be used in several healthcare applications.

Possible Applications

1. Hospital screening systems

Patient information could be automatically analysed during hospital visits to identify people at high risk.

2. Primary healthcare

Doctors could use the model as an additional decision-support system during routine health check-ups.

3. Mobile health applications

Patients could enter basic information such as age, BMI and glucose level to obtain an initial risk assessment.

4. Preventive healthcare

High-risk individuals could be advised to undergo additional clinical testing or modify their lifestyle.

Challenges in Industry

Data Availability

Large, high-quality and accurately labelled medical datasets are difficult to obtain because patient data is sensitive.

Privacy

Healthcare systems must comply with privacy and data-protection requirements. Patient information should not be exposed during model training or deployment.

Model Generalization

A model trained using one hospital's data may perform poorly in another hospital because of differences in:

- Patient population
- Equipment
- Data collection procedures
- Disease prevalence

Computational Requirements

Basic classification models require relatively low computational resources. However, large-scale healthcare AI systems may require significant storage and computing infrastructure.

Ethical Issues

The model should support doctors rather than completely replace them. Incorrect predictions can potentially result in unnecessary testing or delayed treatment.

TASK 4: REFLECTION AND LEARNING OUTCOME

(a) Three Key Insights

Insight 1: Learning from Examples

I learned that **inductive learning** allows a machine-learning model to derive general rules from specific training examples.

For diabetes prediction, the model observes many patient records and learns relationships between clinical features and diabetes outcomes.

Connection:

Inductive Learning

Insight 2: Importance of Feature Selection

I understood that different patient attributes do not contribute equally to prediction. Features such as **glucose level, BMI and age** can provide important information for identifying diabetes risk.

Selecting useful features can improve model efficiency and reduce unnecessary complexity.

Connection:
Decision Trees / Statistical Learning

Insight 3: Accuracy Alone Is Not Enough

The article helped me understand that evaluating a healthcare ML model requires more than accuracy. Metrics such as **precision, recall, F1-score and ROC-AUC** provide a better understanding of classification performance.

For medical diagnosis, recall is especially important because failing to identify a diabetic patient can have serious consequences.

Connection:
Statistical Learning and Model Evaluation

(b) Proposed Novel Extension

Proposed Experiment: Explainable Diabetes Prediction Using XGBoost + SHAP

I would extend the research by developing an **explainable machine-learning model using XGBoost combined with SHAP (SHapley Additive exPlanations)**.

The experiment would use a larger and more diverse diabetes dataset obtained from multiple healthcare institutions. The data would be divided using stratified k-fold cross-validation. XGBoost would then be compared with Decision Tree, Random Forest and Logistic Regression.

SHAP would be used to explain individual predictions.

For example, the system could provide:

Predicted diabetes risk: High

Major contributing factors: High glucose level, high BMI and age.

Feasibility

This experiment is feasible because:

- XGBoost is widely available through Python libraries.
- SHAP provides established explainability methods.
- Public healthcare datasets can be used for initial experiments.
- The models can be trained using ordinary computing resources for moderate-sized datasets.

Expected Impact

The enhancement could improve both **prediction performance and interpretability**.

Explainability is particularly important in healthcare because doctors need to understand why an AI system produces a particular prediction before considering it in clinical decision-making.

